Lab #4: DHCP Starvation

Setup:

- Create an additional machine using the `clone` option of VBox
- Install a DHCP server: `sudo apt-get install isc-dhcp-server`
- Configure the server:
  - `/etc/network/interfaces` for the static network configuration
  - `/etc/dhcp/dhcpd.conf` for the daemon configuration
  - Restart `iface`
  - Test your server to see that it is configured correctly (allow your first machine to connect to the same network - "internal network" in VBox network configurations)

Tasks:

- Write a DHCP starvation attack tool
  - The minimal options are given below:

```
usage: DHCPStarvationNEW.py [-h] [-i IFACE] [-t TARGET]

DHCP Starvation

optional arguments:
  -h, --help            show this help message and exit
  -i IFACE, --iface IFACE
                        Interface you wish to use
  -t TARGET, --target TARGET
                        IP of target server
```

Added challenge (bonus 20%):

- Add persistence to the tool
  - The program must keep the starvation going until the attacker `Break`s the attack:

```
usage: DHCPStarvationNEW.py [-h] [-p] [-i IFACE] [-t
TARGET]

DHCP Starvation

optional arguments:
  -h, --help            show this help message and exit
  -p, --persist         persistant?
  -i IFACE, --iface IFACE
                        Interface you wish to use
  -t TARGET, --target TARGET
                        IP of target server
```

  - Notice the added `--persist` flag in the help dump

Pointers:

- Your DHCP server might not initially work because it might be configured with two (or more) default gateways.
  - You can see if this is the case using `route`
  - Delete one (or more) of the defaults, until you're left with only one
    - `sudo route del default` (or some other variant of the command to erase specific routes)
- VBox might be running a DHCP server even for internal network which you might need to turn off (it's off by default… but go know ☺).
  - `VBoxManage dhcpserver remove --netname <YourInternalNetwork>`
- You should look into running separate threads for the sniffer and the flooder (not required, but will drastically help).

Deliverables:

- A zipped file containing:
  - Your code (.py)
  - A Document explaining
    - The main parts of your code
    - The file from your server machine showing all the active connections at a specific point in time
      - `/var/lib/dhcp/dhcpd.leases`
      - **Please filter the file to include only current active leases (current = time of printout)**

Note:

- You may NOT download or copy code from internet !!
- Searching the internet with specific programming question IS allowed.